Q4

a) By applying First Order Taylor Approximation we have,

$M_{t+1} = f(F_{t+1}, \theta) \approx f(E_t[F], \theta) + \partial_F f(E_t[F], \theta)'(F_{t+1} - E_t[F]) = M_{t+1}^{\text{linear}}$

Since $M_{t+1} = R_{w,t+1}^{-\gamma}$

$F_{t+1} = R_{w,t+1}$ $\theta = \gamma$

$f(F, \gamma) = F^{-\gamma}$

subsuming the actual facts, we have:

$f(E_t[R_{w,t+1}], \gamma) = \left(E_t[R_{w,t+1}]\right)^{-\gamma}$

$f'(E_t[R_{w,t+1}], \gamma) = -\gamma\left(E_t[R_{w,t+1}]\right)^{-\gamma-1}$

$M_{t+1} = R_{w,t+1}^{-\gamma} \approx \left(E_t[R_{w,t+1}]\right)^{-\gamma} - \gamma E_t[R_{w,t+1}]^{-\gamma-1}\left(R_{w,t+1} - E_t[R_{w,t+1}]\right)$

$= \left(E_t[R_{w,t+1}]\right)^{-\gamma} - \gamma E_t[R_{w,t+1}]^{-\gamma-1} R_{w,t+1} + \gamma E_t[R_{w,t+1}]^{-\gamma}$

$= (1+\gamma) E_t[R_{w,t+1}]^{-\gamma} - \gamma E_t[R_{w,t+1}]^{-\gamma-1} R_{w,t+1}$

$a_t = (1+\gamma) E_t[R_{w,t+1}]^{-\gamma}$ $\quad b_t = \gamma E_t[R_{w,t+1}]^{-\gamma-1}$

Thus, $M_{t+1} \approx a_t - b_t \cdot R_{w,t+1}$, $a_t = (1+\gamma) E_t[R_{w,t+1}]^{-\gamma}$ $\quad b_t = \gamma E_t[R_{w,t+1}]^{-\gamma-1}$

For $\gamma > 0$, $a_t - b_t \cdot R_{w,t+1} < 0$ if

$(1+\gamma) E_t[R_{w,t+1}]^{-\gamma} - \gamma E_t[R_{w,t+1}]^{-\gamma-1} R_{w,t+1} < 0$

$(1+\gamma) E_t[R_{w,t+1}]^{-\gamma} < \gamma E_t[R_{w,t+1}]^{-\gamma-1} R_{w,t+1}$

$R_{w,t+1} > \dfrac{(1+\gamma) E_t[R_{w,t+1}]^{-\gamma}}{\gamma E_t[R_{w,t+1}]^{-\gamma-1}} = \dfrac{1+\gamma}{\gamma} E_t[R_{w,t+1}]$

Thus, the linear SDF is not guaranteed to remain positive, unlike the original non linear SDF.

This is a disadvantage because strict positivity of an SDF is sufficient to ensure no-arbitrage pricing.

the linear approximation does not necessarily preserve this property.